



Parsing Biological Text Files

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Institute for Applied Research in Health Science and Aging, Thong Nhat Hospital

Table of Contents

1. Introduction to NCBI

1.1 Introduction to PubMed

1.2 Introduction to GenBank

2. Exercise in Google colab

NCBI

(National Center for Biotechnology Information)

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

All Databases

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

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Popular Resources

- PubMed
- Bookshelf
- PubMed Central
- BLAST
- Nucleotide
- Genome
- SNP
- Gene
- Protein
- PubChem

NCBI News & Blog

BankIt Submitters: GenBank Submission Update 26 May 2026

As previously announced, major changes are being made to enhance your 21 May 2026

Now Available: RefSeq Release 235 21 May 2026

RefSeq release 235 is now available online and from the FTP site! You can access RefSeq data

New Data Available! Access Hantavirus Sequences at NCBI 20 May 2026

Sequence data from the recent Andes hantavirus outbreak are now available

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Website:

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Established in 1988 as a division of the National Library of Medicine (NLM) at the NIH

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Error blocked for possible abuse **Server** misuse.ncbi.nlm.nih.gov **Client** 2001:ee0:1b42:8c8:db57:5d6e:20f1:f232 **Time** Friday, 29-May-2026 22:14:42 EDT

 cửa hàng chrome trực tuyến

 Tìm kiếm tiện ích và giao diện

Khám phá Tiện ích Giao diện



VPN Miễn Phí - Free VPN - 1VPN

 1vpn.org  Nổi bật 4,7 ★ (61,9 N lượt xếp hạng)   Chia sẻ

Tiện ích

Quyền riêng tư và bảo mật

2.000.000 người dùng

Thêm vào Chrome



National Library of Medicine
National Center for Biotechnology Information

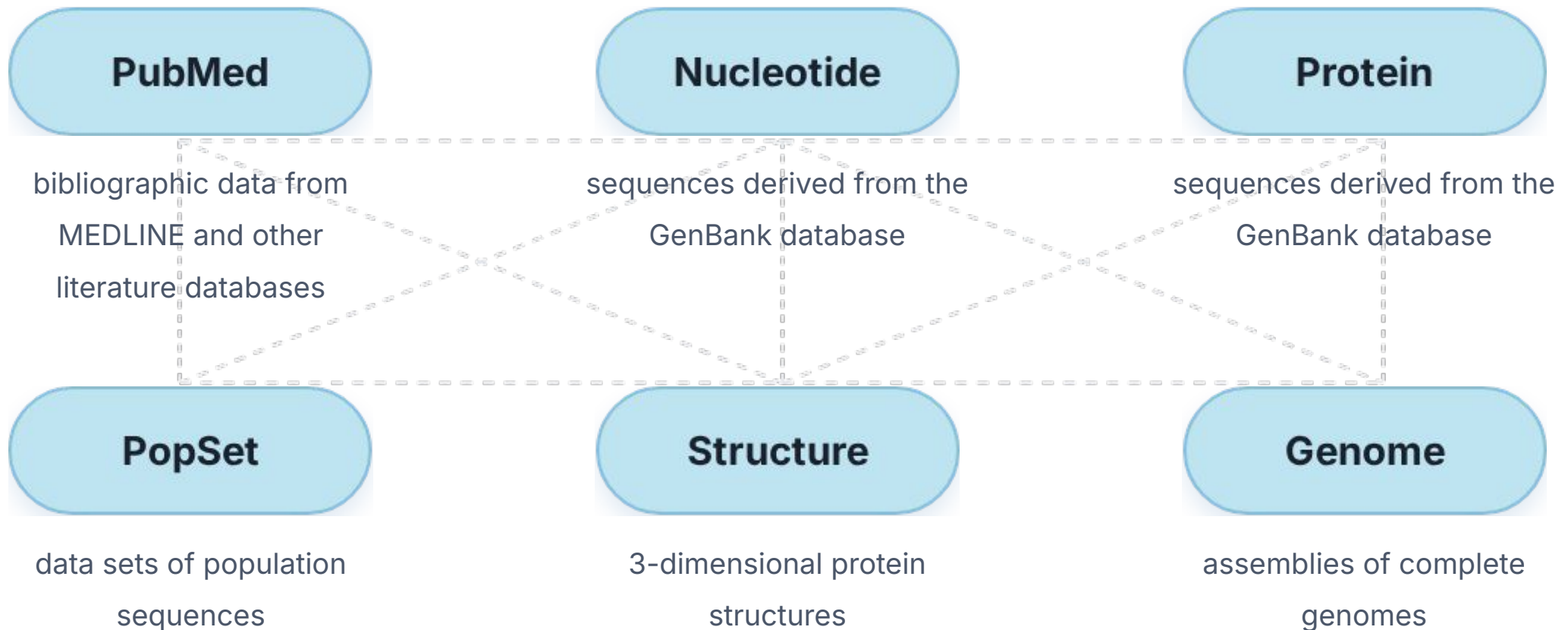
Log in

All Databases ▼

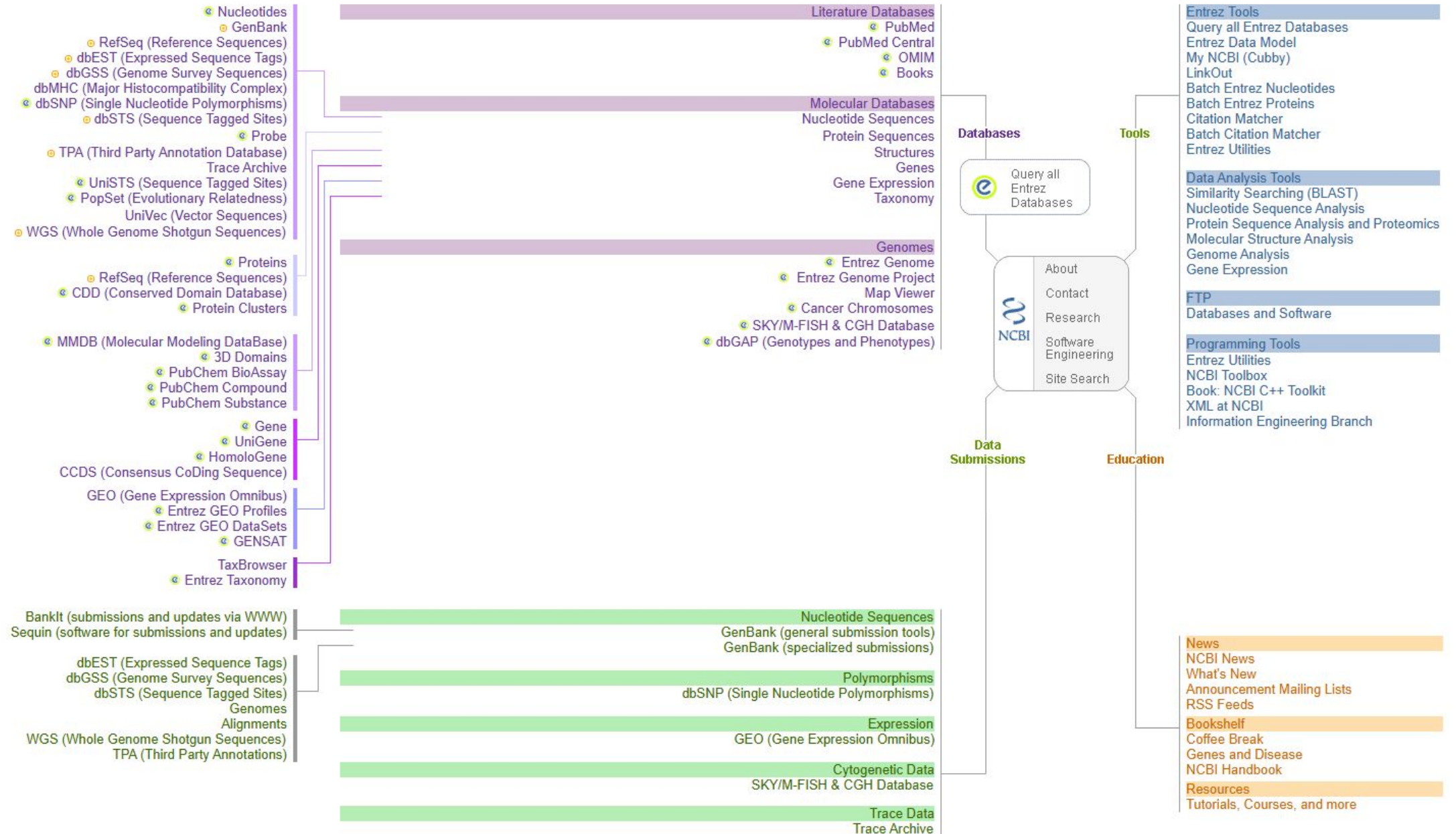
Search

NCBI database component

A unified network enabling comprehensive biological data retrieval.



NCBI site map



NCBI current data scale in 2026

40M
CITATIONS

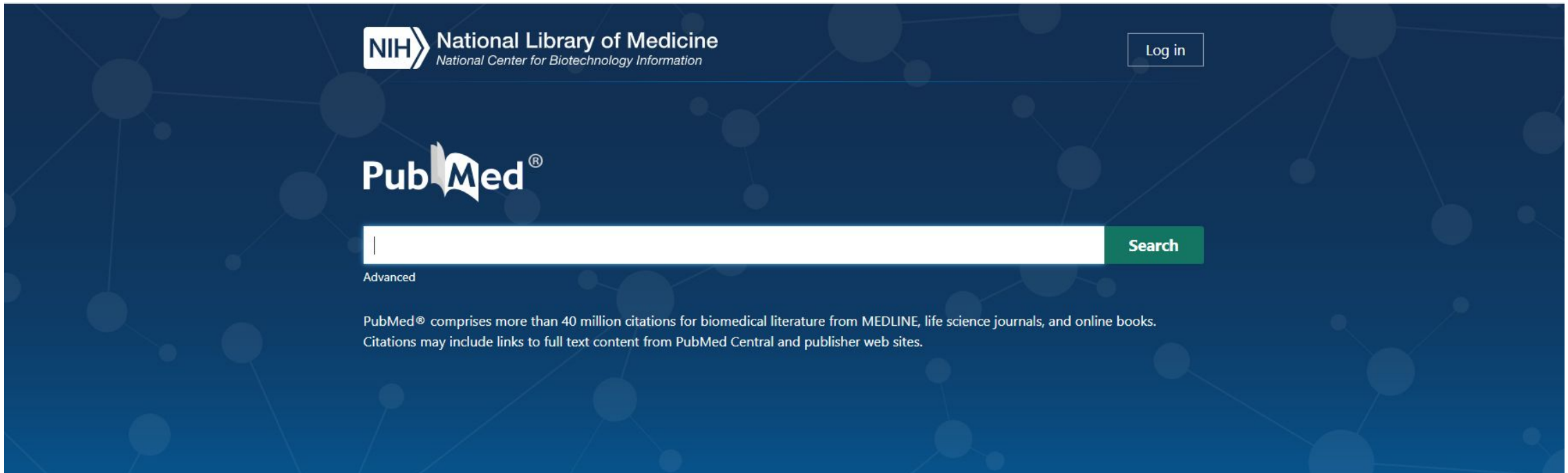
473M
PROTEINS

3.4M
GENOMES

Source: RefSeq Release 235 & NCBI News

PubMed: The Global Gateway to Biomedical Literature.

- **PubMed** is a free, publicly accessible search engine and database developed and maintained by the NCBI
- **Scope:** Medicine, nursing, dentistry, veterinary medicine, the health care system, and preclinical sciences.
- **Full-Text Access:** provides citations and abstracts, it frequently includes links to full-text articles available through **PubMed Central (PMC)** or external publisher websites.



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Explore

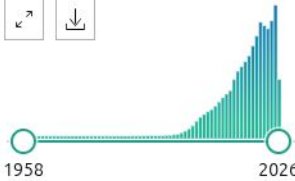
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1958 2026

PUBLICATION DATE

☐ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

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☐ Clinical Trial
☐ Meta-Analysis
☐ Randomized Controlled Trial
☐ Review
☐ Systematic Review

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Sort by: Publication date

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42,499 results

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1958 2026

PUBLICATION DATE

☐ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

☐ Associated data

ARTICLE TYPE

☐ Books and Documents
☐ Clinical Trial
☐ Meta-Analysis
☐ Randomized Controlled Trial
☐ Review
☐ Systematic Review

DISPLAY OPTIONS

- Format: Summary
- Per page: 1
- Abstracts: 1

Filters applied: in the last 1 year, Free full text. Clear filters

Magnetic nanohydroxyapatite-peptide silk fibrous scaffolds for bone regeneration analysis.

1 of 4,250

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Journal

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danph@bvtu.edu.vn

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Filters

Filters for: PubMed

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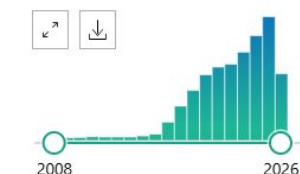
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Edit custom filters

RESULTS BY YEAR



PUBLICATION DATE

- ☐ 1 year
☐ 5 years
☐ 10 years
☐ Custom Range

TEXT AVAILABILITY

- ☐ Abstract
☐ Free full text
☐ Full text

ARTICLE ATTRIBUTE

2,458 results

<< < Page 1 of 246 > >>

- ☐ **Candida auris** Genetics and Emergence.
1 Chowdhary A, Jain K, Chauhan N.
Cite Annu Rev Microbiol. 2023 Sep 15;77:583-602. doi: 10.1146/annurev-micro-032521-015858. Epub 2023 Jul 5.
PMID: 37406342 Free PMC article. Review.
Candida auris is a multidrug-resistant fungal pathogen that presents a serious threat to global human health. Since the first reported case in 2009 in Japan, *C. auris* infections have been reported in more than 40 countries, with mortality rates between 30% and 60%.
- ☐ **Candida auris**: Epidemiology and Antifungal Strategy.
2 Eix EF, Nett JE.
Cite Annu Rev Med. 2025 Jan;76(1):57-67. doi: 10.1146/annurev-med-061523-021233. Epub 2025 Jan 16.
PMID: 39656947 Free PMC article. Review.
Candida auris is a recently emerged fungal pathogen that causes severe infections in healthcare settings around the globe. ...*C. auris* can persist on skin or in the surrounding environment for extended periods of time, and it exhibits greater antifungal resistance.
- ☐ **Comprehensive Overview of Candida auris: An Emerging Multidrug-Resistant Fungal Pathogen.**
3 Kim JS, Cha H, Bahn YS.
Cite J Microbiol Biotechnol. 2024 Jul 28;34(7):1365-1375. doi: 10.4014/jmb.2404.04040. Epub 2024 Jun 15.
PMID: 38881183 Free PMC article. Review.
The rise of **Candida auris**, a multidrug-resistant fungal pathogen, across more than 40 countries, has

PMID (PubMed ID): Assigned to every abstract or citation that enters the standard PubMed database. It only links to the abstract/citation page

Review > Annu Rev Microbiol. 2023 Sep 15;77:583-602.

doi: 10.1146/annurev-micro-032521-015858. Epub 2023 Jul 5.

Candida auris Genetics and Emergence

Anuradha Chowdhary^{1 2}, Kusum Jain¹, Neeraj Chauhan³

Affiliations + expand

PMID: 37406342 PMID: **PMC12962553** DOI: 10.1146/annurev-micro-032521-015858

Abstract

Candida auris is a multidrug-resistant fungal pathogen that presents a serious threat to global human health. Since the first reported case in 2009 in Japan, *C. auris* infections have been reported in more than 40 countries, with mortality rates between 30% and 60%. In addition, *C. auris* has the potential to cause outbreaks in health care settings, especially in nursing homes for elderly patients, owing to its efficient transmission via skin-to-skin contact. Most importantly, *C. auris* is the first fungal pathogen to show pronounced and sometimes untreatable clinical drug resistance to all known antifungal classes, including azoles, amphotericin B, and echinocandins. In this review, we explore the causes of the rapid spread of *C. auris*. We also highlight its genome organization and drug resistance mechanisms and propose future research directions that should be undertaken to curb the spread of this multidrug-resistant pathogen.

Keywords: MAP kinase signaling; amphotericin B; antifungal drug resistance; azoles; cell wall; echinocandin.

[PubMed Disclaimer](#)

PMCID (PubMed Central ID): Assigned *only* when the full-text article is successfully processed and hosted in the PubMed Central repository. It always starts with the letters "**PMC**" followed by a string of numbers.

Exercise

Find in PubMed term: ***oryza sativa*** and answer the questions:

- How many paper free full text published in around 1 year:
 - + Randomized controlled trial?
 - + Review?
 - + Systematic review?
- What is the closely published 3 articles? Their PMID? Give the day of published?
- Save 5 closely published articles as citation text
- Save 1 article to your Bibliography

GeneBank

The annotated collection of all publicly available DNA sequences

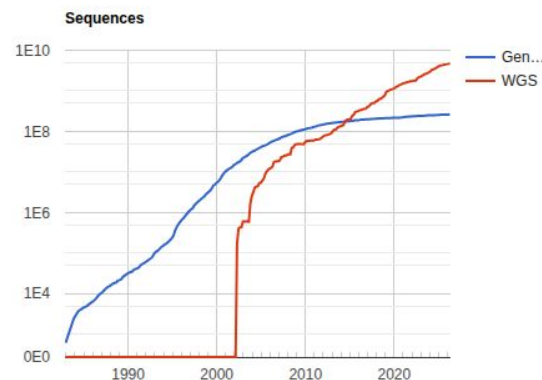
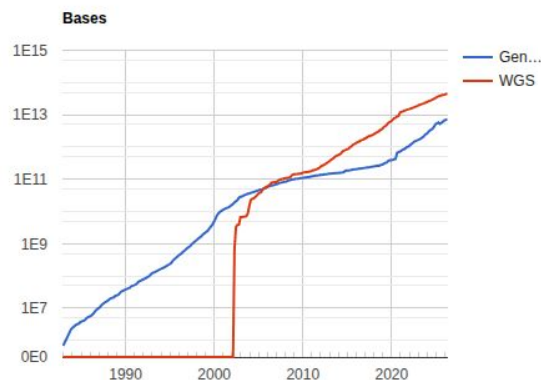
GeneBank update statistic:

<https://www.ncbi.nlm.nih.gov/genbank/statistics/>

GeneBank sample record:

<https://www.ncbi.nlm.nih.gov/genbank/samplerecord/>

GenBank and WGS Statistics



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National Center for Biotechnology Information

GenBank

Nucleotide

GenBank

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Genomes

WGS

Metagenomes

TSA

INSDC

Documentation

Special

GenBank Overview

What is GenBank?

GenBank[®] is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences (*Nucleic Acids Research*, 2013 Jan;41(D1):D36-42). GenBank is part of the [International Nucleotide Sequence Database Collaboration](#), which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI. These three organizations exchange data on a daily basis.

A GenBank release occurs every two months and is available from the [ftp site](#). The [release notes](#) for the current version of GenBank provide detailed information about the release and notifications of upcoming changes to GenBank. Release notes for [previous GenBank releases](#) are also available. GenBank growth [statistics](#) for both the traditional GenBank divisions and the WGS division are available from each release.

An [annotated sample GenBank record](#) for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

Access to GenBank

There are several ways to search and retrieve data from GenBank.

- Search GenBank for sequence identifiers and annotations with [Entrez Nucleotide](#).
- Search and align GenBank sequences to a query sequence using [BLAST](#) (Basic Local Alignment Search Tool). See [BLAST info](#) for more information about the numerous BLAST databases.
- Search, link, and download sequences programatically using [NCBI e-utils](#).
- The ASN.1 and flatfile formats are available at NCBI's anonymous FTP server: <ftp://ftp.ncbi.nlm.nih.gov/ncbi-asn1> and <ftp://ftp.ncbi.nlm.nih.gov/genbank>.

GenBank Data Usage

The GenBank database is designed to provide and encourage access within the scientific community to the most up-to-date and comprehensive DNA sequence information. Therefore, NCBI places no restrictions on the use or distribution of the GenBank data. However, some submitters may claim patent, copyright, or other intellectual property rights in all or a portion of the data they have submitted. NCBI is not in a position to assess the validity of such claims, and therefore cannot provide comment or unrestricted permission concerning the use, copying, or distribution of the information contained in GenBank.

Data Processing, Status and Release

The most important source of new data for GenBank is direct submissions from a variety of individuals, including researchers, using one of our submission [tools](#). Following submission, data are subject to automated and manual processing to ensure data integrity and quality and are subsequently made available to the public. On rare occasions, data may be removed from public view. More details about this process can be found on the [NLM GenBank and SRA Data Processing](#).

GeneBank format

Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence

GenBank: AB243863.1

[FASTA](#) [Graphics](#)

Go to:

LOCUS AB243863 1504 bp DNA linear BCT 08-DEC-2010
DEFINITION Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence.
ACCESSION AB243863
VERSION AB243863.1
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SOURCE Bacillus sp. YM9-109
ORGANISM [Bacillus sp. YM9-109](#)
Bacteria; Bacillati; Bacillota; Bacilli; Caryophanales;
Bacillaceae; Bacillus.
REFERENCE 1
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Novel Bacillus species isolated from soil and sea water
JOURNAL Published Only in Database (2010)
REFERENCE 2 (bases 1 to 1504)
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Direct Submission
JOURNAL Submitted (06-DEC-2005) Contact:Keiichi Goto Mitsui Norin Co.Ltd.,
Food Reseach Laboratories; 223-1, Miyabara, Fujieda, Shizuoka,
426-0133 JAPAN, Fujieda, Shizuoka 426-0144, Japan
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301 ctgagacacg gccagagctc ctacgggagg cagcagtagg gaatttcca caatggacga
361

GeneBank's information components:

- Nomenclature and Classification
- Sequence Information
- Accession and Versioning
- Genome/Assembly Details
- Modification History
- Associated Publication

LOCUS AFB76520 207 aa linear INV 22-JUL-2013
DEFINITION COI, partial (mitochondrion) [Escarpia sp. 4 JMP-2012].
ACCESSION AFB76520
VERSION AFB76520.1
DBSOURCE accession [JN021270.1](#)
KEYWORDS .
SOURCE mitochondrion Escarpia sp. 4 JMP-2012
ORGANISM [Escarpia sp. 4 JMP-2012](#)
Eukaryota; Metazoa; Spiralia; Lophotrochozoa; Annelida; Polychaeta;
Sedentaria; Canalipalpata; Sabellida; Siboglinidae; Escarpia.
REFERENCE 1 (residues 1 to 207)
AUTHORS Raggi,L., Schubotz,F., Hinrichs,K.U., Dubilier,N. and Petersen,J.M.
TITLE Bacterial symbionts of Bathymodiolus mussels and Escarpia tubeworms
from Chapopote, an asphalt seep in the southern Gulf of Mexico
Environ. Microbiol. 15 (7), 1969-1987 (2013)
JOURNAL
PUBMED [23279012](#)
REFERENCE 2 (residues 1 to 207)
AUTHORS Raggi,L., Schubotz,F., Hinrichs,K.-U., Petersen,J.M., Felden,J. and
Dubilier,N.
TITLE Direct Submission
JOURNAL Submitted (25-MAY-2011) Symbiosis group - Molecular Ecology, Max
Planck Institute for Marine Microbiology, Celsiusstr. 1, Bremen,
Bremen 28359, Germany
COMMENT Method: conceptual translation.
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CDS 1..207
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Nucleotide [Help](#)

GenBank

Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence

GenBank: AB243863.1

[FASTA](#) [Graphics](#)

[Go to:](#)

LOCUS AB243863 1504 bp DNA linear BCT 08-DEC-2010
DEFINITION Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence.
ACCESSION AB243863
VERSION AB243863.1
KEYWORDS .
SOURCE Bacillus sp. YM9-109
ORGANISM [Bacillus sp. YM9-109](#)
Bacteria; Bacillati; Bacillota; Bacilli; Caryophanales;
Bacillaceae; Bacillus.
REFERENCE 1
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Novel Bacillus species isolated from soil and sea water
JOURNAL Published Only in Database (2010)
REFERENCE 2 (bases 1 to 1504)
AUTHORS Goto,K., Asahara,M., Kasai,H. and Yokota,A.
TITLE Direct Submission
JOURNAL Submitted (06-DEC-2005) Contact:Keiichi Goto Mitsui Norin Co.Ltd.,
Food Reseach Laboratories; 223-1, Miyabara, Fujieda, Shizuoka,
426-0133 JAPAN, Fujieda, Shizuoka 426-0144, Japan
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/collection_date="Nov-2003"
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rRNA
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/gene="rrs"

Send to:

☒ Complete Record
☐ Gene Features

Choose Destination

☒ File ☐ Clipboard
☐ Collections ☐ Analysis Tool

Download 1 item.

Format

Show GI ☐

Send to => Complete Record =>
File => FASTA => Create File

FASTA format

```
>AB243863.1 Bacillus sp. YM9-109 rrs gene for 16S rRNA, partial sequence
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```

>Header line: Begins with a greater-than (>) symbol in the first column

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NFITTVMMNRYKGLRLERVPLFVWSVKLTAVLLLLSIPVLAGGLTMLLTDRNLNTSFFDPAGGGDPV
```

Sequence Data: The actual nucleotide (e.g., A, C, G, T) or amino acid sequence data.

IUPAC (International Union of Pure and Applied Chemistry)

Code	Represents	Complement
A	Adenine	T
G	Guanine	C
C	Cytosine	G
T	Thymine	A
Y	Pyrimidine (C or T)	R
R	Purine (A or G)	Y
W	weak (A or T)	W
S	strong (G or C)	S
K	keto (T or G)	M
M	amino (C or A)	K
D	A, G, T (not C)	H
V	A, C, G (not T)	B
H	A, C, T (not G)	D
B	C, G, T (not A)	V
X/N	any base	X/N
-	Gap	-

Exercise

Find the accession number and answer the questions:

- What is the name of gene?
- What is the species name?
- Length of sequence?
- Molecule type? (Genomic DNA, genomic RNA, precursor RNA, mRNA (cDNA), ribosomal RNA, transfer RNA, small nuclear RNA, or small cytoplasmic RNA?)

1. MW854701.1

3. MZ005317.1

5. AB087252.1

2. AB039789.1

4. CM131255.1

6. AH009855.2

